

3-B:

→ The proposal was not socially beneficial which means that Social cost is greater than Social Benefit.

B - Since private benefit is \$900m, and external benefit is \$300m, the social benefit will be \$1200(PB+EB)

Now, we know that, social costs must be greater than \$1200(which is the social benefit.)

We know that private costs are 800, so now in-order for SC to be greater than 1200, the external costs must be greater than 400. This is why, key B, becomes correct.

A – If external costs were equal to 300(EB), then SC will not be greater than SB.

C – External costs(>400) should be greater than External benefits(=300), in order for social costs to be greater than social benefits.

D – If there were no external costs, then Social Benefits would be greater than Social costs, and in that case, the project would not have been rejected.

4-B:

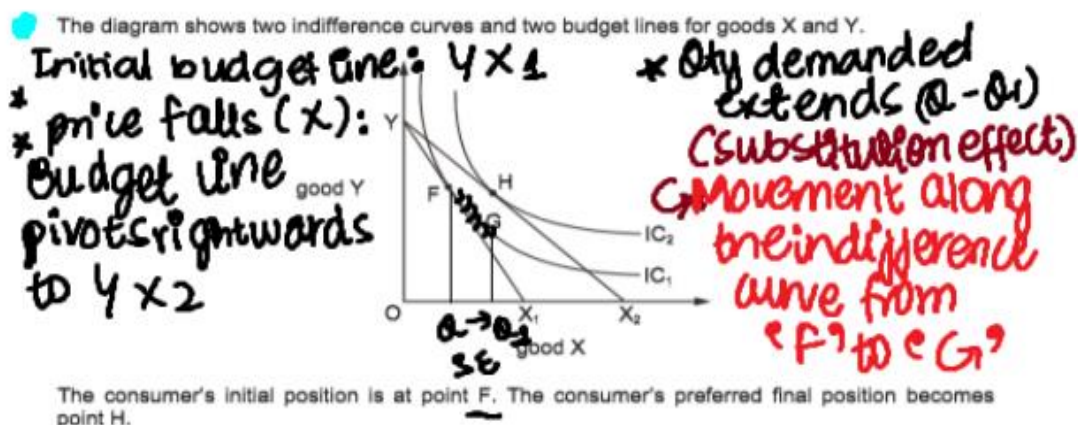
A – Total utility only falls when marginal utility becomes negative.

B – Because of a fall in the price, there is an extension in qty demanded of the good, which brings about a movement along the “demand curve”, which is why, this becomes correct

C – Because of an extension in qty demanded as a result of a fall in the price of the good, there is a fall in the marginal utility derived from consuming the good

D -A shift in the demand curve occurs as a result of other factors, and not because of the “price factor”

5-C:



A- The income effect of a price fall for X would be from “G to H”

B- The price effect represents the total effect, which comprises of both: income effect and the substitution effect, and is shown from “F” to “H”

C- It can be seen in the diagram above that the substitution effect for a fall in the price of X is shown through the movement along the indifference curve from “F to G”

D- This is incorrect as the quantity would have contracted if the price had risen for X

6-B:

A: Concentration of activities in which a country possesses “comparative advantage” explains the concept of “ international specialization”, which is macro-related.

B: Vertical integration involves “forward-vertical integration” and “backward-vertical integration”. Because of backward integration, the firm has a contract with **the raw material or the component provider**, and the availability of components at a cheaper rate helps to cut costs. Moreover, because of forward-vertical integration, the **service sector is under the firm, or in other words, the firm has access to the retail outlet at low cost** , and as a result of which, the firm is again able to undercut costs. This is why, key B, becomes correct.

C: An increase in the market share is earned because of horizontal integration, that is when firms in the same industry merge at the same stage of production.

D: Improvement in efficiency resulting from increased use of market incentives is also related to “horizontal integration”

7-A:

A - The scenario described, where the organizers of a major sporting event produce official souvenir products and street traders sell cheaper unofficial souvenirs, is an example of a contestable market. A contestable market is characterized by low barriers to entry and exit, allowing new firms to enter and compete with existing firms, which is why, the potential threat of competition always exists. In this case, the street traders are able to enter the market and offer cheaper alternatives to the official souvenirs, providing competition to the organizers.

B: Perfect competition: This option is incorrect. Perfect competition refers to a market structure where there are numerous buyers and sellers (1000's of producers), homogeneous products, perfect information, and free entry and exit. While the presence of street traders may introduce competition, the scenario does not explain perfect competition.

C : Price discrimination : This option is incorrect. Price discrimination refers to the practice of charging different prices to different customers for the same product or service based on the price elasticity of demand, which also determines their willingness to pay. The scenario does not explicitly mention different prices being charged based on customer characteristics or willingness to pay.

D: Price leadership : This option is incorrect. Price leadership occurs when one dominant firm in an industry sets the price, and other firms follow suit. The scenario does not indicate any firm setting the price and others adjusting accordingly. This is practiced by a “dominant oligopoly”

8-C:

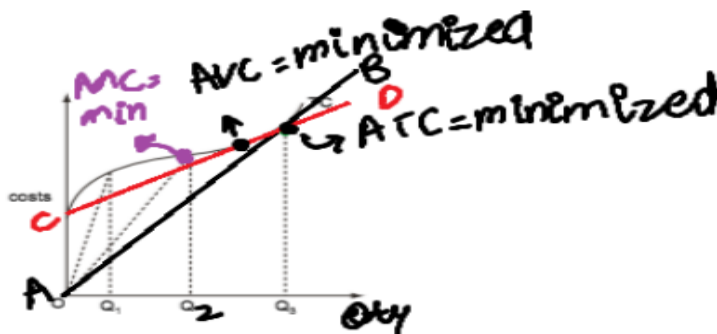
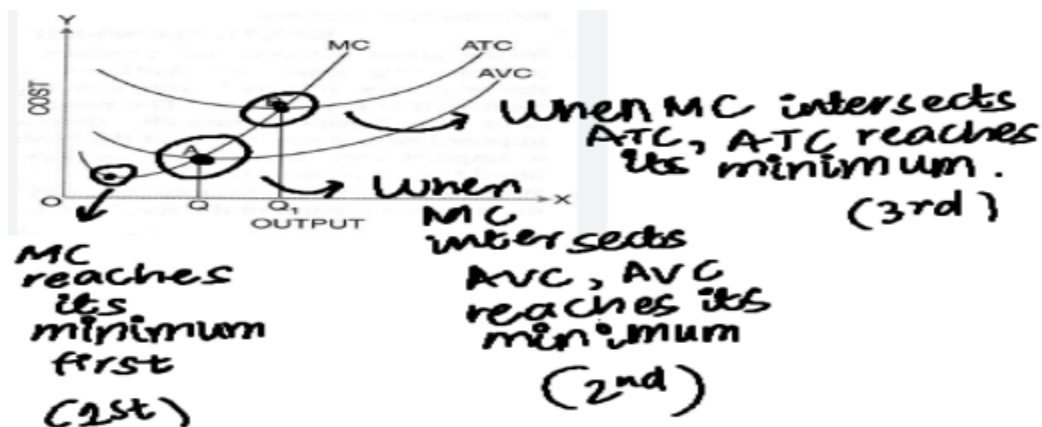
A: While calculating a firm’s short-run production at-least one factor of production is assumed to be fixed which is capital, and **labor is assumed to be variable**

B: All factors of production are variable in the “long-run” and not the “short-run”

C: This is correct as the state of technology is assumed to be fixed when calculating the short-run production function

D: The state of technology is variable in the “long-run”

9-D:



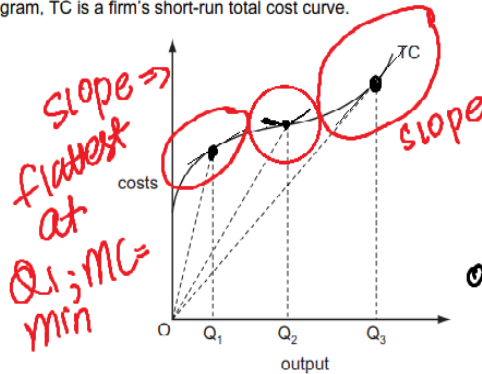
A – At “OQ3”, ATC is minimized since the line from the origin(AB) is tangent to the total cost curve there.

B- It can be seen above that the average variable cost is minimized between output Q2 and Q3. This is because the line below the total cost curve (CD) is tangent to the total cost curve there.

C- It can be seen above that the average variable cost is minimized between output Q2 and Q3. This is because the line below the total cost curve (CD) is tangent to the total cost curve there.

D: At Q2, marginal cost is at its minimum, this is because, the total cost curve is the flattest there, which means that the change in total cost or the marginal cost is at its lowest. This is why, key D, becomes correct

In the diagram, TC is a firm's short-run total cost curve.



• To find out the slope of the TC curve, we need to draw lines (corresponding to each level of output) that are tangent to the TC curve.

• Where the slope is the flattest, that is when, the marginal cost or the change in total cost is at its minimum.

10-A:

At zero level of output, total cost is "40". This represents "fixed cost", since at output of zero, only fixed cost is incurred.

At output level of 4, the total cost is 80, and after subtracting "40", that is the fixed cost, we will obtain "40" as the variable cost.

Average variable cost is calculated by dividing Total Variable Cost by Qty

It is : $40/4$ i.e. "10"

This is why, key A, becomes correct.

11-D:

A – It is in a perfectly competitive market structure that all firms are **price-takers**, because their market share is insignificant to influence the price level.

B – All firms in the industry producing an identical or homogeneous product represents a "perfectly competitive market structure"

C – There are a large number of producers (there are 1000s of them) in case of "perfect competition"

D- In case of a 'perfectly contestable market', there are zero barriers to entry and exit. A common example used to illustrate a perfectly contestable market is the market for ice cream trucks. In this hypothetical scenario, there are no substantial barriers to entry or exit, such as licensing requirements or exclusive territorial rights. New ice cream truck vendors can easily enter the market, offer competitive prices and products, and attract customers. Existing vendors, meanwhile, can exit the market without incurring significant costs or restrictions.

12-C:

The diagram shows the average and marginal revenue curves of an oligopolistic firm. OQ is the profit-maximising output.

Which curve could be the firm's marginal cost curve?



→ The **Marginal cost curve** should be in the range where Marginal revenue curve is dis-continuous, because only then, the corresponding price will be the one unchanged at the 'kink'.

→ Therefore, the **marginal cost curve** should be 'C' because, it must lie **between the discontinuous MR change**, because it reflects the fact that the price is set at the 'kink'.

Options A and B reflect the fact that the price will be set in **the elastic zone**.

→ If a firm in a '**non-collusive oligopoly**' raises price in an attempt to raise Total Revenue, other firms will not follow, and therefore, this behavior of the firm is categorized under '**elastic demand**', and since **quantity demanded contracts by a relatively higher percentage**, this will lead to a fall in "**Total Revenue**"

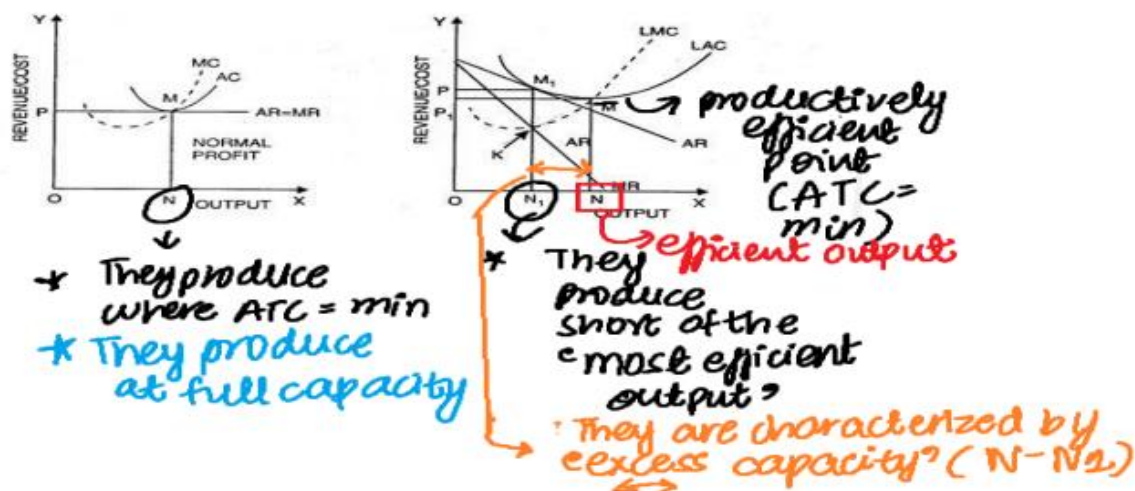
Option D represents that the price is in the inelastic zone.

→ If a firm in a '**non-collusive oligopoly**' decreases price in an attempt to raise Total Revenue, **other firms in the industry will follow**, and since there is no such alternative for the industry, this represents **inelastic demand**, and in response to this, quantity demanded extends by a relatively smaller percentage, which represents the fact that **Total Revenues** will fall.

This is why, key C, becomes correct.

13-A:

A – In contrast to a perfectly competitive firm, monopolistically competitive firms are characterized by “excess capacity”, which means that they produce short of “somewhat the most efficient output”. The long-run equilibrium positions of monopolistic competition, and perfect competition are shown below.



B – This is incorrect because, both of these market structures make “normal profit” in the long-run.

C – This is incorrect because, monopolistic competition has a smaller number of firms (mainly 20-25 small firm) unlike perfect competition, which has a large number of firms (as 1000's of producers operate)

D- This is incorrect as the “demand curve of perfect competition is perfectly elastic”, as compared to the demand curve of “monopolistic competition”, which is highly elastic. (It is relatively “less elastic than perfect competition”)

14-B:

A: Imperfect information is a source of market failure -> If individuals are not aware of the benefits or positive externalities associated with consuming merit goods, they may not prioritize their consumption, even if it is in their best interest or contributes to the overall welfare of society. For example, preventive healthcare services or education may be underutilized if individuals are not fully informed about their long-term benefits. This under-allocation of merit goods results in market failure.

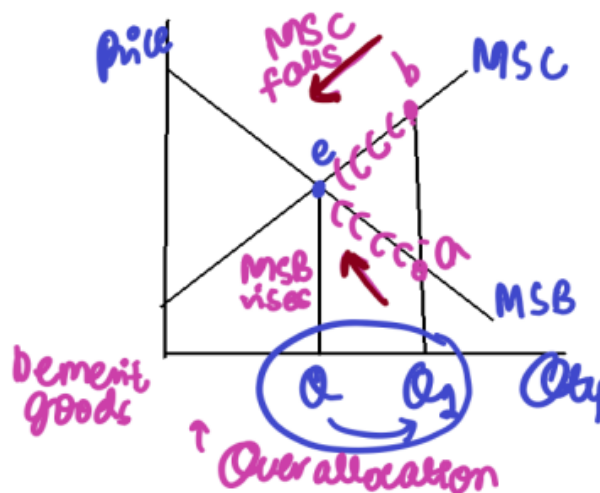
-> If individuals are unaware of the harmful consequences or negative externalities associated with demerit goods, they may over- consume them without fully considering the long-term repercussions. For example, individuals may engage in excessive consumption of harmful substances such as: alcohol, due to a lack of information about the health risks involved. This over -allocation of de-merit goods results in market failure.

B) Income Inequality: Income inequality refers to the unequal distribution of income among individuals or households in an economy. This is a macro-economic concern and may not cause market failure. This is why, key B becomes correct.

C) Monopoly is a source of market failure. A “monopoly” occurs when a single firm dominates the market and restricts competition. In a monopoly, the monopolistic firm has significant market power, allowing it to set prices higher than in a competitive market, resulting in reduced output and consumer welfare. This results in “market failure”.

D) Non-excludability is a source of market failure, particularly concerning public goods. Public goods are goods or services that are non-excludable, meaning that payment by one individual does not restrict others from enjoying the benefits associated with these goods; even the non-tax payers get to consume it. This results in the free-rider problem. Due to the free-rider problem, where individuals can benefit from public goods without paying for them, private markets do not provide for them, and this non-provision of public goods, results in market failure.

15-D:



A- This is incorrect because MSB rises when quantity is contracted.

B- Again, Marginal social benefit rises when quantity is contracted.

C - If the fall in the total social benefit is greater than the total social cost, this is a disadvantage, which will result in a fall in "society's net welfare"

D - If the fall in total social cost is greater than the fall in the total social benefit, this represents a "fall in external cost", as a result of which, society's net social welfare increases.

16-A:

A) Changing a universal benefit to a means-tested benefit is the least likely to help a household out of the poverty trap. Means-testing can create a poverty trap, where individuals are discouraged from working or increasing their income due to fear of losing benefits. Not only do individuals lose benefits suddenly when their income exceeds a certain threshold, but also they are liable to pay tax.

B) Increasing the threshold at which income tax is paid can help households out of poverty by allowing them to keep more of their earnings. This can be especially beneficial for low-income households, who may see a significant increase in their take-home pay.

C) Introducing a national minimum wage can help households out of poverty by ensuring that low-income workers earn a living wage. This can help them meet their basic needs and potentially escape poverty.

D) By reducing the tax burden, individuals may be more likely to enter the workforce or increase their hours worked, thereby helping them escape poverty. This explains why this won't contribute to a poverty trap.

17-C:

A – Marginal cost of labor does not represent the addition to the revenue as a result of employing one additional unit of labor.

B – This is incorrect because it is “marginal revenue productivity” which represents the addition to the revenue as a result of employing one additional unit of labor. **And, marginal revenue is only a part of Marginal Revenue Productivity, which is why, option B, becomes incorrect.**

C – This is correct because **marginal revenue productivity** represents the **additions to the revenue** as a result of **employing one additional unit of labor**. **An additional unit of labor will contribute to the output as a result of which, it contributes in form of “additions to the total revenue”**. This is why, key C, becomes correct.

D – This is incorrect because the law of diminishing returns sets in when the Marginal Product of Labor falls.

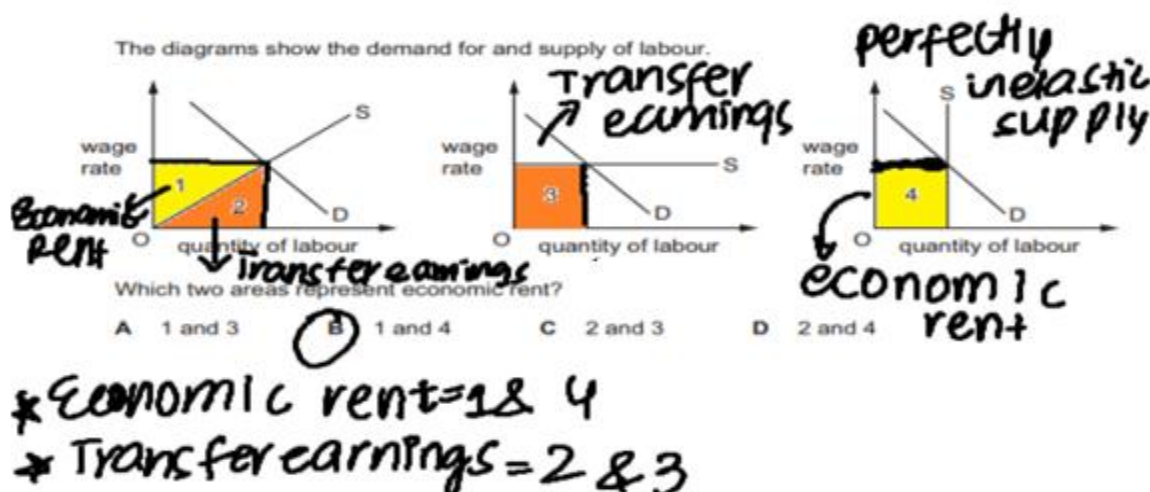
18-B:

Transfer earnings are the minimum income a worker needs in order to supply their labor.

Economic rent is the extra income a worker receives – above the minimum level they need in order to work.

→ The area below the supply curve of labor represents “transfer earnings”.

→ The area above the supply curve of labor and below the wage rate represents the “economic rent”



The diagram below explains why, key B, will be correct.

19-B:

The table shows the short-run production function of a firm with labour as the variable factor of production.

Labour cost →

* wage X no of workers employed.

wage	number of workers employed	total output per day	labour cost	labour cost / output
50	1	10	50	5
50	2	20	100	5
50	3	40	150	3.75
50	4	50	200	4
50	5	60	250	4.2

→ lowest labour cost per unit of output

If workers are hired in a perfectly competitive labour market at a wage rate of \$50 per day, when will the labour cost per unit of output be at its lowest?

☐ A 2 workers
☒ B 3 workers
☐ C 4 workers
☐ D 5 workers

20-C:

- A)** An increase in overtime means that employment is unlikely to decrease, as existing employees are already working more hours.
- B)** Overtime can lead to higher earnings, which can drive inflation, not reduce it. As workers earn more, they have more disposable income to spend, potentially increasing consumption, Aggregate demand and prices when Aggregate demand outpaces Aggregate Supply.
- C)** This is the correct answer. When overtime increases, average earnings grow faster than hourly wage rates. This is because overtime pay is often at a premium rate so total earnings increase more rapidly than the base hourly rate.

-> The increase in overtime will lead to an increase in Total Earnings, which will in turn increase Average Earnings (Total Earnings / Number of Employees). However, the Hourly Wage Rate might not increase as much, since the premium overtime rate is not reflected in the regular hourly rate.

By using the formula, we can see how the increase in overtime leads to a more rapid growth in average earnings than in hourly wage rates, making option C the correct answer.

- D)** It may also be that an increase in output exceeds the increase in labors worked, thereby resulting in an increase in labor productivity.

21-B:

Underemployment is a common feature of developing economies, where workers may be employed in low-productivity jobs or may work fewer hours than they would like. This can be due to a lack of job opportunities arising as a result of limited economic growth. **This explains why key B will be correct.**

- A)** A relatively small agricultural sector is not typical of developing economies, which often have a large agricultural sector.

- C)** High saving rates are not characteristic of developing economies, which often have low saving rates due to average income levels.

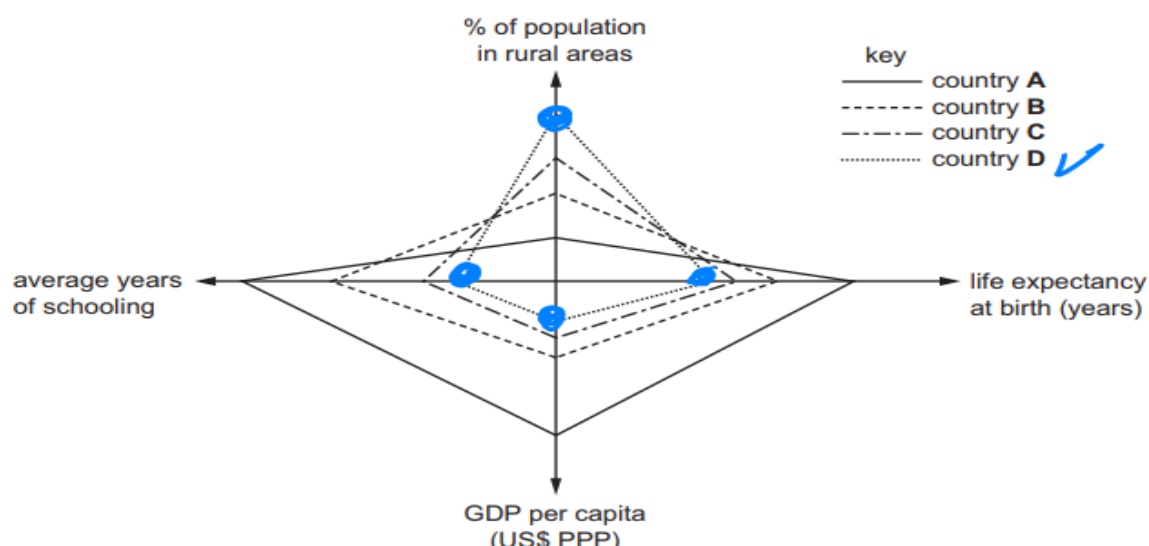
D) Stable export earnings are not typical of developing economies, which often face volatility in export earnings due to dependence on primary commodities and fluctuations in global prices.

22-D:

In a least developed economy:

- The percentage of population living in rural areas should be the highest.
- Life expectancy at birth should be the lowest.
- Average years of schooling should be the lowest.
- GDP per capita should be lowest.

Which country is likely to be **least** developed?



23-D:

③ $mpc + mps = 1$

↓ if it's greater than unity ↓ mps will be -ve

④ $1.25 + (-0.25) = 1$

> 1

Marginal Propensity to Consume (MPC): The proportion of additional income that is spent on consumption.

Marginal Propensity to Save (MPS): The proportion of additional income that is saved.

If $MPC > 1$, it means that :-

MPS must be negative, as the sum of MPC and MPS is always equal to 1 (D is the best answer)

A) Since option A is linked with Average propensity to consume, this becomes incorrect.

- When consumption is greater than income, it represents dissaving.
- This means APC (Average Propensity to Consume) is >1 and APS (Average Propensity to Save) is negative.
- $APC + APS = 1$

B) Inflation might occur if aggregate demand increases, but it's not a direct consequence of $MPC > 1$.

C) A trade deficit might occur if consumption includes imports, but it's not a necessary consequence of $MPC > 1$.

24-D:

The private sector surplus (Saving - Investment) represents the net leakages

because:- If Savings (leakages) $>$ Investment(injections), the excess savings are leaking out of the circular flow. This represents net leakages from the circular flow.

Government budget deficit (No): When government spending (injections) are greater than tax revenues (leakages), this represents net injections into the circular flow.

Trade surplus (No): When export revenues (injections) are greater than import expenditures (leakages), this represents net injections into the circular flow.

25-B: The formula of marginal propensity to consume is: $\text{Change in consumption} / \text{Change in income}$.

Disposable Family Income (\$)	Consumption Expenditure (\$)	Savings (\$)	Change in Consumption (\$)	Change in Income (\$)	MPC
2000	2150	-150	-	-	-
3000	3100	-100	950	1000	0.95
4000	4000	0	900	1000	0.9
5000	4850	150	850	1000	0.85
6000	5650	350	800	1000	0.8
7000	6380	620	730	1000	0.73

26-B:

Transactions Demand for Money:

The transactions demand for money refers to the demand for money for everyday transactions, such as buying goods and services.

Factors that Increase Transactions Demand for Money:

- An increase in nominal national income (B) leads to more transactions, as people have more money to spend, increasing the demand for money.

A) A fall in the price of bonds indicate a situation whereby interest rates are high since there is a negative relationship between interest rates and the price of bonds. High interest rates would result in an increase in savings, because of which, consumption may decrease. This results in a decrease in transactionary demand of money. This is because, there is a negative relationship between interest rate and quantity demanded of money.

C) An increase in the rate of interest would result in an increase in savings, because of which, consumption may decrease. This will consequently result in a decrease in transactionary demand of money.

D) An increase in unemployment would likely lead to a decrease in transactionary demand of money, as people have less income to spend.

27-D:

The most important line over here is, “government economic advisers disagree over the relative impact of fiscal and monetary controls”

Let’s take an example of two economists, who are having a discussion on which policy measure to adopt in event of inflation. Economic adviser A, since is in favor of the monetarists, prefers that monetary policy is more effective, and is therefore, trying to argue on the likely ineffectiveness of contractionary fiscal policy, which is preferred by economic adviser B, who seems to favor Keynesian economics. Since, they are continuously having disputes, there will be a time lag that will occur “in selection of the appropriate policy”. Therefore, option D, becomes correct.

Note: The question asks about a time lag that might occur after it is recognised that there is a need for government economic advisers. Option C comes before that stage. The order would then be, select a policy (option D), introduce the policy to the situation (option A), monitor its effectiveness (option B).

28-B: **Why is B the correct answer?**

- International trade has not decreased, but rather increased, with the rise of globalization and international agreements like WTO and regional trade agreements.

The other options are incorrect because they have accompanied global economic growth over the last 25 years:

- A) Depletion of non-renewable resources has occurred due to increased energy demand.
- C) Growing urbanization has led to increased economic activity and growth.
- D) Increased atmospheric pollution has resulted from growing industrial activity and energy consumption.

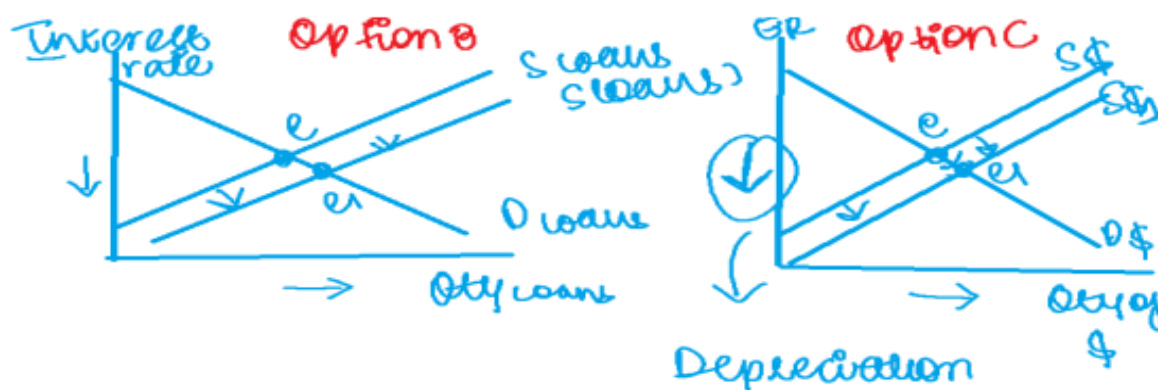
29-D:

A – A reduction in interest rates are a part of an expansionary monetary policy, as it will stimulate investments and Aggregate demand.

B- A reduction in the minimum cash to deposit ratio of banks, means that since commercial banks are obliged to hold less cash in liquid form, they can give out more of the deposits in form of loans, and when the supply of loans rises, it results in a reduction in interest rates. A reduction in interest rates are a part of an expansionary monetary policy, as it will stimulate investments and Aggregate demand. (ref. graph below)

C- If the central bank purchases foreign currency after selling the domestic currency, since this would increase the supply of currency, it would result in a fall in the exchange rate(representing “depreciation”), which will lead to a Bot surplus, thereby helping to stimulate AD; this again represents expansionary monetary policy.(ref. graph below)

D – Now, since the central bank sells government’s securities in the open market, this since results in a decrease in money supply relative to money demand, would lead to a rise in interest rates. This will have a contractionary effect on AD, and so, this option becomes correct.



30-B:

The principal aim of fiscal policy (option B) is to ensure sound public finances, since it is also known as a budgetary policy, which means that through this, it is decided to either operate a deficit or a surplus. This is why, key B becomes correct.

